

The Entropy Ledger: Operational Limits for Consistent Time Travel

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Summary. We present operational limits for retrocausal resources: Loop-DPI bounds loop information gain by $-\log p_{\text{succ}}$ bits per attempt; a thermodynamic Entropy Ledger prices paradox resolution at $kT D_{\text{KL}}$. Single-shot converses and multi-loop cut-sets are provided, with experimental proposals and an operational metaphysics framing.

Main

We formalize a heralded retro-map within a CPTP circuit. The Loop-DPI stems from a Radon–Nikodym bound on the success-conditioned measure. The Entropy Ledger follows from I-projection and nonequilibrium work relations. Empirically, heralded photonics and ion-trap protocols can chart the Pareto between temporal advantage and entropy debt.

Figures

See Extended Data: empirical-vs-bound, Pareto frontier, multi-loop scaling, single-shot smoothed bounds, QuTiP heatmap.

Methods

Derivations, proofs, and simulation details are in the Appendices.

Data and Code Availability

All simulations and $\text{\LaTeX}$ sources are included.

References